

Cross compiling Linux kernel for RPi3 (Model B)

First, install the build dependencies:

```
$ sudo apt-get -y install git fakeroot build-essential ncurses-  
dev xz-utils libssl-dev bc flex libelf-dev bison libgmp3-dev
```

1. Download the RPi toolchain in your Ubuntu machine from the given website:

Command to download toolchain in ubuntu machine:

```
$ cd (Jump to home directory)  
$ git clone https://github.com/raspberrypi/tools
```

Toolchain can be downloaded directly from the given website

Website: <https://github.com/raspberrypi/tools>

2. Download Linux kernel source code for RPi3 modelB.

Note: 5.12.19-v71 Linux version is preferable

Website: <https://github.com/raspberrypi/linux/tree/rpi-5.12.y>

3. Export given environment variables in ubuntu machine. Use the given commands to do it.

```
$ export ARCH=arm  
$ export CROSS_COMPILE=arm-linux-gnueabihf-  
$ export PATH=$PATH:/home/<login_name>/tools/arm-bcm2708/arm-  
linux-gnueabihf/bin
```

4. Kernel compilation steps for RPi3B

```
$ cd /<path>/linux-rpi-5.12.y  
$ make bcm2711_defconfig (select RPi3B model)  
$ make (this will take some time)
```

5. After compilation compiled kernel image is available in given director of linux kernel source code [Name of the kernel image is "zImage"]

```
/<path>/linux-rpi-5.12.y/arch/arm/boot
```

6. Kenel porting:

Connect memory card where Raspbian os is installed. Then enter to bootfs (FAT32) partition of the memory card. Copy zImage to kernel7.img.

Or

Another way is, login to rpi. Mount /dev/mmcblk0p1 partition in /tmp directory in rpi. Then transfer zImage from your laptop/pc to rpi using scp command. Then, in the rpi, copy zImage to kernel7.img in /tmp directory. Unmount the /tmp partition.

Syntax of scp command:

```
sudo scp /path_of_file/filename login_name_of_pi@ip_addr_of_pi:/dest_path_in_rpi
```

eg:

```
sudo scp zImage pi@192.168.100.10 :/home/pi
```

- 7 . Reboot your rpi after replacing the kernel image.
8. Connect SD card to RPi & bootup your pi.
9. Type uname -r command to cross verify kernel porting.